

Google Summer of Code 2015 Proposal

Elasticsearch-lua client

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= BASICS =

1) What is your preferred e-mail address?

me@dhavalkapil.com

2) Do you have a web page, blog, github?

Github: <https://github.com/dhavalkapil>

Blog: <https://dhavalkapil.com/>

PS: I also contribute blogs here: <https://blog.sdslabs.co/>

3) What is your academic background?

I'm a second year undergraduate Computer Science student at IIT Roorkee enrolled in a 4 year course.

4) What other time commitments, such as school work, another job, planned vacations, etc., will you have between May 25 and August 21?

I'm completely free from May 25 to July 14. I have no other commitments/jobs. Apart from 2-3 days vacations(which I will let you know in advance) I shall work everyday. I can easily target about 50 hours a week.

After July 14 I have regular college. I might spend a little bit less time during weekdays but I promise to make it up during weekends. Overall I'll be able to give 35-40 hours a week.

= EXPERIENCE =

1) What programming languages are you fluent in?

Which tools do you normally use for development?

I am fluent in Java, C, C++, PHP, Lua, Javascript, Nodejs. Otherwise I can also program in python, ruby and bash(but require some googling here and there).

I use both linux(ubuntu 14.04) and windows on my laptop. For development I prefer linux unless the language is microsoft specific(eg. C#). I am quite familiar with the terminal and hardly use my mouse. I use git as the version control system. I prefer to use editor+terminal rather than IDE for development. Sublime Text is my favourite editor. It makes development quite fast. For minor edits in files I use vim.

Apart from that, for Android applications I use Eclipse IDE and for C# I use Visual Studio for development.

2) Are you familiar with the Lua programming language?

Have you developed any project using Lua?

I am quite familiar with the lua language. I know about the fundamentals involving tables, metatables, etc.

I have worked on a project known as Brute-DC. It basically is a client that scrapes file on the DC protocol and provides a web interface for easy searching. I have written lua scripts for it that are run on a nginx server.

I've also written a small prototype of an elasticsearch client in lua. Code is here:

<https://github.com/DhavalKapil/elasticsearch-lua>

3) Have you developed software in a team environment before?

Yes, I have a good experience in developing software in a team environment. I am a member of SDS Labs, an awesome group of students constantly striving to innovate and foster technical activities at IIT Roorkee. Some of our projects are hosted on github:

<https://github.com/sdslabs/>. I have contributed to many of their projects(both public and private) along with many other developers. My experience as being part of SDS Labs can be seen here: <https://blog.sdslabs.co/2014/12/working-sdslabs-6/>

Apart from that I have participated in hackathons in teams of 2,3 and 4. We used git to collaborate. I've even been one of the winners of Microsoft Code.Fun.Do. Code for my entry is hosted here: <https://github.com/DhavalKapil/harley>.

4) a) What kinds of project/software have you worked on previously? (anything larger than a class project: academic research, internships, freelance, hobby projects, etc.)

I've worked on many project/software:

(Team projects)

Backdoor (<https://backdoor.sdslabs.co/>)

Backdoor is a platform for hackers to show their talent in a competitive environment. Earlier it was launched only within the IIT Roorkee campus, but now it has been made available for anyone over the internet. It is a Capture the Flag styled event. A flag (basically a string) is associated with every challenge. Once you have solved a particular challenge you would get the hidden flag which you need to submit back to us for getting points.

As part of SDS Labs I developed the backend for this platform. I also contribute challenges.

This application won me the first prize under the category of 'Dynamic website' in Srishti IITR(Annual technical exhibition of IIT Roorkee).

Brute-DC (presently not available on the internet)

It is basically a client that scrapes file on the DC protocol. It provides a web interface for in depth searching. I have written lua scripts concerning the DC protocol part of the project. They are run

on a nginx server. This project will soon be launched as soon as the web interface part is finished by other developers.

JEE listing (<https://jee.sdslabs.co/>)

I alongwith a few members of SDS Labs got the JEE 2014 Ranklist and released it. We were the first(and perhaps the only) people to release it unofficially. It's code is open sourced:

<https://github.com/sdslabs/jee>

Muzi (available only on the intranet of IITR)

Muzi is perhaps the most famous application on the intranet of IITR. It is an online music player. It was developed a few years ago. Due to the always increasing size of the database the 'search' feature of the application had become quite slow. Recently I implemented elasticsearch for searching of songs in the backend using elasticsearch-php client. Using it increased the performance tremendously!

Harley

It's a Kinect based application to help autistic children with mobility impairment and shape selection problems. I developed it as part of Microsoft Code.Fun.Do hackathon. We were the runners up.

<https://github.com/DhavalKapil/harley>

(Personal projects)

dns-validator (<https://www.npmjs.com/package/dns-validator>)

It's a security tool to detect dns poisoning attacks. I developed this while learning NodeJs.

Source code at: <https://github.com/DhavalKapil/dns-validator>

network-monitor (<https://www.npmjs.com/package/network-monitor>)

It's a simple tool that analyses and monitors HTTP traffic over a particular interface and displays results in a TUI based responsive window. It's written in NodeJs. Source code at:

<https://github.com/DhavalKapil/network-monitor>

viewer

A client/server java application to transmit changes on files on the client side to the server side immediately. This is basically used to synchronize between two directories on different machines. Source code at: <https://github.com/DhavalKapil/viewer>

elasticsearch-lua

I developed a small prototype client for elasticsearch in lua with basic operations including searching, indexing and deleting a document. It currently just uses a single connection to a node. The final client would obviously be much more complex(as specified in the architecture and design I provided).

Source code at: <https://github.com/DhavalKapil/elasticsearch-lua>

**b) Have you ever developed software for "real users"?
(i.e., other than yourself)**

Yes I have. I mentioned backdoor previously. Currently more than 1000 users from all over the world are registered. It was just launched in March on the internet. Muzi is used extensively on the intranet of IITR. Other projects such as dns-validator and network-monitor also are being used by different users(as seen by the stats on their respective websites).

c) In particular, are you (or have you been) involved with any open source development project? If so, briefly describe the project and the scope of your involvement.

Some of the projects mentioned above are open sourced. As such I have not contributed in some famous organization but I do contribute little bit in other organizations(as seen through my github profile).

= PROJECT =

1) Did you select a project from our list?

If yes, which project did you select? Why did you choose this project?

If you are proposing a project, please give a description of your proposal, including expected results.

Yes I selected elasticsearch-lua as the project that I wish to apply.

Reason -> I have experience with elasticsearch. I have worked on a project involving indexing chat messages and searching them using mongodb+elasticsearch. I even have implemented elasticsearch on one of my projects. I have a fair idea about the backend of elasticsearch including working of shards, nodes and clusters. I have also used elasticsearch-php extensively in one of my projects. Apart from that I also have experience in network based applications involving sockets. Moreover I am quite interested in elasticsearch and want to develop it further.

As far as lua is concerned I have written many scripts for it. I am aware of the power of the various features of this language.

Proposal - elasticsearch-lua:

Basically I would like to create a **low level** client in lua for elasticsearch. Taking inspiration from other elasticsearch clients and personal experience I would suggest a modular architecture for the elasticsearch client. It would consist of the following modules:

1. **Connection** - This module will basically deal with the connection to a particular node in the cluster. After searching and experimenting for a while I'd prefer luasocket for connecting with the elasticsearch server. The main reason is that they provide a nice interface for HTTP requests(GET, POST, PUT, DELETE) that are required to interact with the server. As far as multiple connections are concerned, it won't be effective unless we perform batch requests. Mostly the

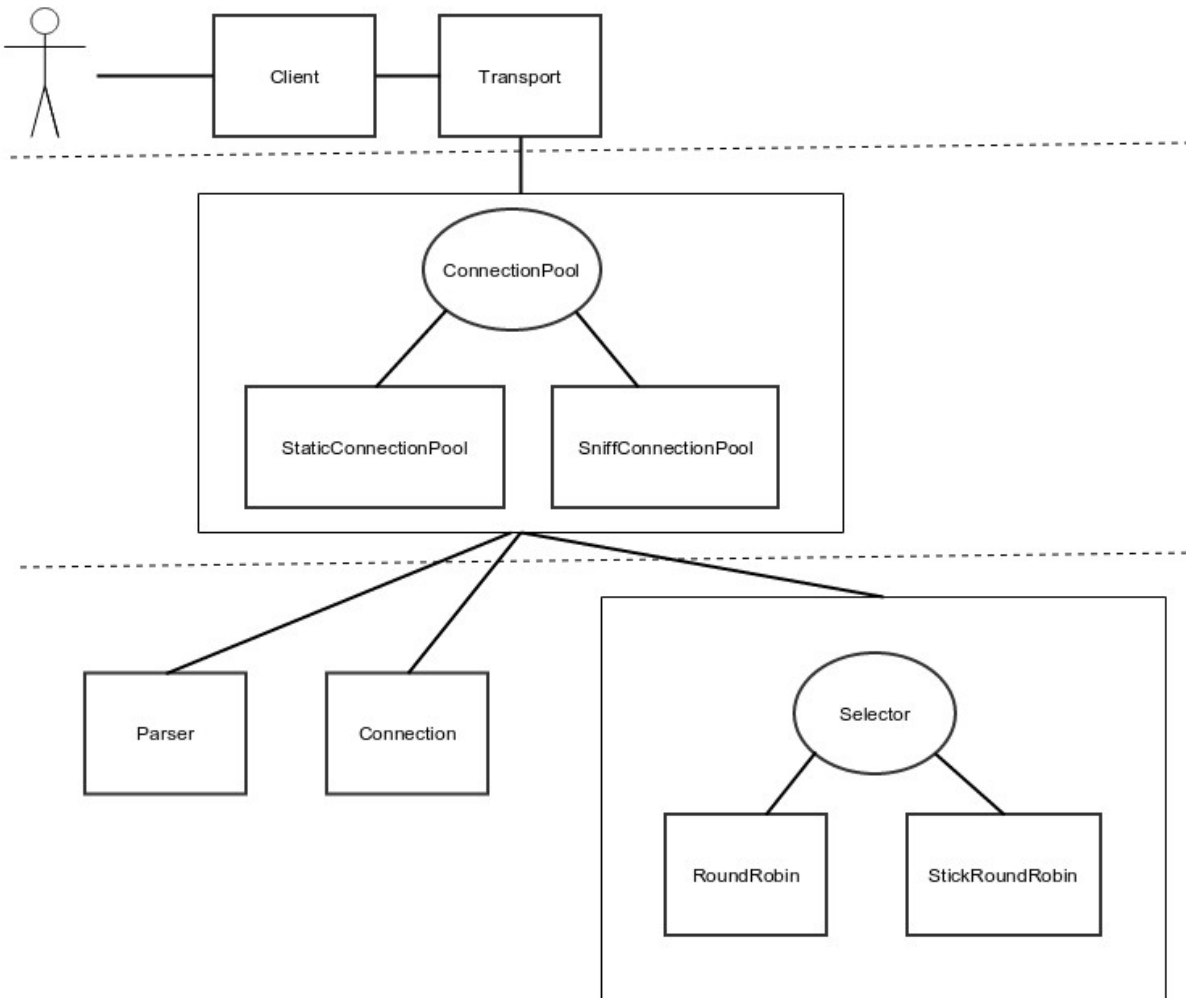
requests will be single. Even though I'd add support for it by using multiple threads. Lua Lane library should be used as it will optimize performance on multi-core CPU's and would decrease the effective time during multiple requests.

2. **Selector** - This module will select the next connection from the list of all connections depending on various factors. I would like to add support for various algorithms. The user will be given a choice to choose one.
 - a. **Round Robin(default)** - The connections would be selected in a simple round robin fashion. Every time a new connection is asked for, the next connection in the list will be returned. After all connections are selected at least once the list would be started once again. In this way all of the connections would be given equal priority.
 - b. **Stick Round Robin** - This will be same as simple Round Robin except for the fact that one connection will be reused till it dies away. After that the next connection in the list will be selected. Will need to experiment with both to determine the most optimum.

Note that the Selector just selects a new connection mathematically and does not deal with whether the connection is alive or not.

3. **ConnectionPool** - This module will deal with the list of all connections to the different nodes of a cluster. This will actually select a connection with the help of selector. After the selector selects one connection we can have the following implementations(again customizable by the user):
 - a. **StaticConnectionPool(default)**: This is perhaps the simplest implementation. It will test the connection returned by the selector. Depending upon it's status(alive or dead) it will returned or else the selector will again be called.
 - b. **SniffConnectionPool**: Using the initial seed list provided by the user, we use the Cluster State API provided by elasticsearch to discover all the rest nodes. This will be dynamic in the sense that the list will be constantly updated depending on the nodes that are active.
4. **Parser** - The module will deal with the parsing of the input/output. The user will send the input in the form of a lua table which will need to be converted to proper json format to be passed to the elasticsearch server. Also the response would be in json format and would need to be converted to lua table. I would prefer MP-CJSON/LUA-CJSON because of it's extremely fast speed for encoding and decoding.
5. **Transport** - This module will be a layer over the ConnectionPool and Parser. It will provide a new connection for querying the elasticsearch server and would handle errors and retries.
6. **Client** - This module will be at the topmost layer. The user will interact with the elasticsearch client using this module. It will provide an in depth API for performing basic operations like search, index, delete, bulk, count, get, scroll, etc. Apart from that API would also be provided to deal with various nodes and shards.

The architecture and the interconnections of the various modules would be like:



2) Please provide a schedule with dates and important milestones/deliverables (preferably in two week increments).

April 27 to May 25	Discussing and improving the design of the client. Through experience I have learnt that designing a project is a very important part of any software project. Will also study some other clients(php, js) in detail.
May 26 to June 8	Finalizing the design of the client and setting up basic infrastructure and architecture of the application. All the modules would be finalized and skeleton code shall be written.

June 9 to June 16	Keeping the basic requirements for Transport and Client in mind, will develop the two lower layer modules: Connection and Selector along with documentation.
June 16 to June 25	Continuing with the requirements I will write code for the Parser and ConnectionPool. This documentation and the unit tests will also be completed.
June 26 - July 3	Mid term evaluation, discussion with mentor and tweaking existing codebase.
July 4 - July 12	Finishing up Code for Transport module.
July 13 - July 25	Will develop the Client module along with minor add ons to lower layer modules. Documentation and unit testing will be done side by side.
July 26 - August 1	Fixing any bugs, adding minor features left out during initial phases. Integration testing. Discussion with mentor. I expect a working module to be ready by the end of this period.
August 2 - 16	Adding documentation for the user of the API, minor refactor for better code base, more bug fixing.
August 17 - 25	Polishing of the final product with the consultancy of the mentor.

Details on Timeline:

Task 1:

Going through other elasticsearch clients, taking inspiration from them and finalizing a skeleton design for my client.

Task 2:

Start developing the lower layer modules taking into consideration about the features required by top layer modules. Most of the code for Connection, Selector(all of them) and Parser will be written along with documentation and unit test.

Task 3:

Start developing the modules of the upper layer - ConnectionPool(all of them) and Transport again with documentation and testing. Some features will be added to lower layers to implement full functionality. All the while everything will be reported/discussed with the mentor.

Task 4:

Developing the Client module with a full fledged API to match with other clients. It will require learning about other elasticsearch clients.

Task 5:

Integration testing and bug fixing to make everything 'work'. It will be followed by a proper documentation so that it is easier for other developers to use elasticsearch-lua.

Task 6:

The last task is to always polish up the project. There are always a few things that are left during the development phase. This phase is the main difference between a project that is ready to be launched or not.

Note: Basically I aim to develop the lower layer modules first(keeping in mind the functionality required by the module layer over it). While developing the upper layer there would be some minor changes in the lower layer also to accommodate each and every feature.

3) What will be showable two months into the project?

Two months into the project I expect a working module to be ready. Almost all of the features related to documents like indexing, searching, retrieval, scrolling, counting, bulk operations, etc. will be supported. Also API function calls for dealing with nodes, shards, etc. would be working. There will be minor bugs that would be left along with instructions for using this library.

4) Why I should be selected for GSOC?

I'm an enthusiastic developer. I have a rich experience in working on team projects. Also I've known the importance of the various phases of software engineering. While developing anything I do stress upon the design phase and testing phase. Also I am a quick learner. I takes me very little time to go through a particular documentation and use it's API.

I document code while writing it. It really helps the future development process.

I am known to meet hard deadlines. Some of the projects that I have worked on spanned around for 4-5 months. I don't drop a project once I take it.

As mentioned above I already have a lot of experience with lua and elasticsearch. I promise to adhere strictly to the deadlines mentioned in the above table. I know my capabilities perfectly and have formulated the timeline accordingly. I know when to ask for help and when not to.

This project will be a very good learning opportunity for me. It will be my first contribution to such a big organization. I would communicate with my mentor every week and will discuss with him my progress, objectives or any issue that I am facing.